Student Name: _____ Mark: _____ / 39

Moles, molar mass and energy changes: practice test

Time allowed: 50 minutes Total: 39 marks

Instructions

Answer every question in the spaces provided.
You may use a calculator.
Write the numbers into the equation before you calculate. Show your working.
The numbers in the answer boxes are in the units shown next to them.

Data and formulae you may use

Molar mass (atomic mass, Ar) in g mol^{-1}. The element symbol is in the top row. Its molar mass is directly beneath it.

H	C	N	O	Na	Mg	Al	S	Cl	Ca	Cu	Zn
1	12	14	16	23	24	27	32	35.5	40	64	65

Molar mass of a compound: add up the molar masses of all the atoms in its formula.

Moles from a mass

$$n = m \div M$$

Mass from moles

$$m = n \times M$$

n = moles in mol, m = mass in g, M = molar mass in g mol^{-1}

Moles of a solution

$$n = \text{concentration} \times \text{volume} \div 1000$$

concentration in mol dm^{-3}, volume in cm^3

Energy transferred

$$q = v \times 4.2 \times \Delta t$$

q in J, v = volume of water in cm^3, Δt in °C. If two solutions are mixed, v is the total volume. 1 cm^3 of water has a mass of 1 g. 1 kJ = 1000 J.

Enthalpy change

$$\Delta H = q \div n$$

q in kJ, n in mol, ΔH in kJ mol^{-1}. Temperature rise: exothermic, ΔH negative. Temperature fall: endothermic, ΔH positive.

1 System and surroundings [6 marks]

Citric acid is added to sodium hydrogencarbonate solution in a beaker. A thermometer in the solution shows that the temperature falls.

1.1 Which is the system? Tick one box.

- ☐ The citric acid and the sodium hydrogencarbonate
- ☐ The beaker and the thermometer
- ☐ The air in the room [1]

1.2 Which are the surroundings? Tick one box.

- ☐ The citric acid and the sodium hydrogencarbonate
- ☐ Only the citric acid
- ☐ The beaker, the water, the thermometer and the air [1]

1.3 Circle the correct words. Energy flows
from the system to the surroundings / from the surroundings to the system [1]

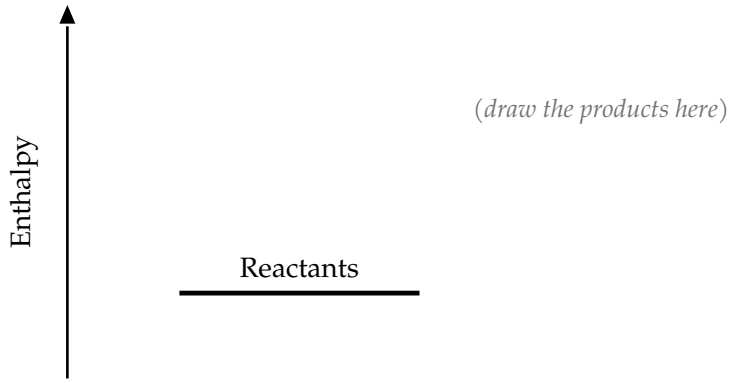
1.4 Circle the correct word. The reaction is exothermic / endothermic. [1]

1.5 Circle the correct word. The sign of ΔH is negative / positive. [1]

1.6 The surroundings lose 20 kJ of energy. Energy is conserved. Circle the correct answer. The energy of the system
decreases by 20 kJ / increases by 20 kJ / does not change [1]

2 Enthalpy level diagrams [8 marks]

2.1 The reactants level of an endothermic reaction is drawn below. Draw the products level on the diagram and label it Products.



[1]

2.2 Draw an arrow on the diagram to show ΔH, and label it ΔH. [1]

2.3 Complete the sentence. For this reaction ΔH is negative / positive because the products are lower / higher than the reactants. (Circle one word in each pair.) [1]

2.4 Figure 1 shows the enthalpy levels for a different reaction. Read the enthalpy of the reactants from the scale.

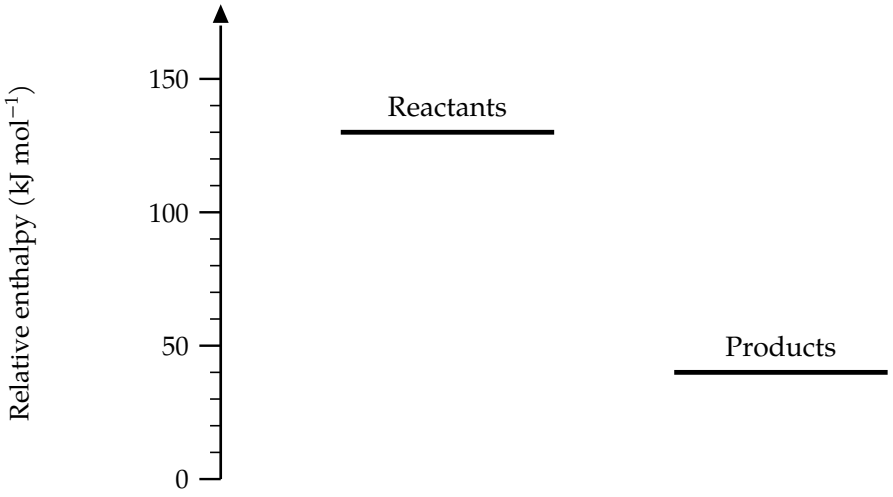


Figure 1: the scale is relative enthalpy in kJ mol^{-1}

Enthalpy of the reactants = ☐ kJ mol^{-1} [1]

2.5 Read the enthalpy of the products from the scale.

Enthalpy of the products = ☐ kJ mol^{-1} [1]

2.6 Work out ΔH for this reaction. Include the sign.

ΔH = ☐ kJ mol^{-1} [2]

2.7 Circle the correct word. The reaction in Figure 1 is exothermic / endothermic. [1]

3 Molar mass and moles [10 marks]

3.1 Complete the table to work out the molar mass of copper(II) sulfate, $CuSO_4$. Use the Ar values on page 1.

Element	Number of atoms	Ar	Number of atoms × Ar
Cu	1		
S	1		
O	4		

[2]

3.2 Add the three numbers from the last column.

Molar mass of $CuSO_4$ = ☐ g mol^{-1} [1]

3.3 Work out the molar mass of H_2SO_4.

Molar mass of H_2SO_4 = ☐ g mol^{-1} [1]

3.4 Calculate the mass of 0.5 mol of H_2SO_4. Use this equation:

$$m = n \times M$$

Write the numbers into the equation first.

$$m = \square \times \square = \square \text{ g}$$ [2]

3.5 Work out the molar mass of SO_2.

$$\text{Molar mass of } SO_2 = \square \text{ g mol}^{-1}$$ [1]

3.6 You have 96 g of SO_2.
Think carefully before calculating. Are you given more than one mole, or less than one mole just looking at the numbers given? Tick one box.

- ☐ More than one mole
- ☐ Less than one mole
- ☐ Exactly one mole

[1]

3.7 Calculate the number of moles in 96 g of SO_2. Use this equation:

$$n = m \div M$$

Write the numbers into the equation first.

$$n = \square \div \square = \square \text{ mol}$$ [2]

4 Empirical formula [5 marks]

6.5 g of zinc reacts with 7.1 g of chlorine to form zinc chloride. Find the empirical formula of zinc chloride.

4.1 Think carefully before calculating. Are you given more than one mole, or less than one mole just looking at the numbers given? Tick one box for each element, then complete the table. Use this equation:

$$\text{moles} = \text{mass} \div \text{Ar}$$

Element	Mass (g)	Ar	More or less than one mole? (tick)	Moles = mass ÷ Ar	Divide by the smallest number of moles
Zn	6.5		☐ more ☐ less		
Cl	7.1		☐ more ☐ less		

Empirical formula: ☐

[5]

5 Energy from a temperature change [10 marks]

30 cm^3 of 1.0 mol dm^{-3} hydrochloric acid is mixed with 20 cm^3 of an excess of sodium hydroxide solution (CARE: CORROSIVE!) in a polystyrene cup. The temperature of the mixture rises from 19.0 °C to 25.5 °C. Treat the mixture as water. The acid is the limiting reactant.

5.1 The two solutions are mixed in the cup. Work out the total volume of the mixture. This is the v you will use in the energy equation.

Total volume = ☐ + ☐ = ☐ cm^3 [1]

5.2 Work out Δt.

Δt = ☐ °C [1]

5.3 Calculate the energy transferred, q, in joules. Use this equation:

$$q = v \times 4.2 \times \Delta t$$

Write the numbers into the equation first.

q = ☐ × 4.2 × ☐ = ☐ J [2]

5.4 Change your answer to kilojoules.

q = ☐ kJ [1]

5.5 Circle the correct word. The reaction is exothermic / endothermic. [1]

5.6 Think carefully before calculating. Are you given more than one mole, or less than one mole just looking at the numbers given? You are given 30 cm^3 of 1.0 mol dm^{-3} hydrochloric acid. Tick one box.

- ☐ More than one mole
- ☐ Less than one mole
- ☐ Exactly one mole

[1]

5.7 Calculate the moles of hydrochloric acid. Use this equation:

$$n = \text{concentration} \times \text{volume} \div 1000$$

n = ☐ × ☐ ÷ 1000 = ☐ mol [1]

5.8 Calculate the size of ΔH in kJ mol^{-1}, with q in kJ.

$$\Delta H = q \div n$$

Size of ΔH = ☐ ÷ ☐ = ☐ kJ mol^{-1} [1]

5.9 Write ΔH with its sign.

ΔH = ☐ kJ mol^{-1} [1]

End of assessment